HARD · ADVANCED MATH · ANSWER KEY

Advanced Math

06

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 A

A. -8

Choice A is correct. The given expression is equivalent to $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$. By combining like terms, this can be written as $9x^4 + x^3 - 8x^2 - x$. This expression is in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. Therefore, the value of a is -8.

Choice B is incorrect. This may result from adding $-5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice C is incorrect. This may result from adding $5x^2$ and $-3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice D is incorrect. This may result from adding $5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Q2 110

The correct answer is 110. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}\left(2x + 22\right)\left(x - 10\right)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}\left(2x + 22\right)\left(x - 10\right) = x^2$. This equation can be rewritten as $\left(x + 11\right)\left(x - 10\right) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding 110 to both sides of this equation yields $x = 110$. Therefore, the value of x is 110.

Q3

D

D. Zero

Choice D is correct. A point x, y is a solution to a system of equations if it lies on the graphs of both equations in the xy-plane. In other words, a solution to a system of equations is a point x, y at which the graphs intersect. It's given that the first equation is $y = 18$. Substituting 18 for y in the second equation yields $18 = -3x - 18^2 + 15$. Subtracting 15 from each side of this equation yields $3 = -3x - 18^2$. Dividing each side of this equation by -3 yields $-1 = x - 18^2$. Since the square of a real number is at least 0, this equation can't have any real solutions. Therefore, the graphs of the equations intersect at zero points.

Alternate approach: The graph of the second equation is a parabola that opens downward and has a vertex at $18, 15$. Therefore, the maximum value of this parabola occurs when $y = 15$. The graph of the first equation is a horizontal line at 18 on the y-axis, or $y = 18$. Since 18 is greater than 15, or the horizontal line is above the vertex of the parabola, the graphs of these equations intersect at zero points.

Choice A is incorrect. The graph of $y = 15$, not $y = 18$, and the graph of the second equation intersect at exactly one point.

Choice B is incorrect. The graph of any horizontal line such that the value of y is less than 15, not greater than 15, and the graph of the second equation intersect at exactly two points.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4

A

A. 4

Choice A is correct. If the given expression is equivalent to $x+b$, then $\frac{x^2-c}{x-b}=x+b$, where x isn't equal to b. Multiplying both sides of this equation by $x-b$ yields $x^2-c=(x+b)(x-b)$. Since the right-hand side of this equation is in factored form for the difference of squares, the value of c must be a perfect square. Only choice A gives a perfect square for the value of c.

Choices B, C, and D are incorrect. None of these values of c produces a difference of squares. For example, when 6 is substituted for c in the given expression, the result is $\frac{x^2-6}{x-b}$. The expression x^2-6 can't be factored with integer values, and therefore $\frac{x^2-6}{x-b}$ isn't equivalent to $x+b$.

Q5

35

The correct answer is 35. It's given that the function $gx = -\frac{1}{55}x + 19x - 35$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. It follows that when the submersible device is at the surface of the ocean, the value of gx is 0. Substituting 0 for gx in the equation $gx = -\frac{1}{55}x + 19x - 35$ yields $0 = -\frac{1}{55}x + 19x - 35$. Multiplying both sides of this equation by -55 yields $0 = x + 19x - 35$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $x + 19 = 0$ or $x - 35 = 0$. Subtracting 19 from both sides of the equation $x + 19 = 0$ yields $x = -19$. Adding 35 to both sides of the equation $x - 35 = 0$ yields $x = 35$. Since $x > 0$, 35 minutes after collecting the sample the submersible device reached the surface of the ocean.

Q6

9

The correct answer is 9. The given equation can be rewritten as $54 - x = 25$. Dividing each side of this equation by 5 yields $4 - x = 5$. By the definition of absolute value, if $4 - x = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting 4 from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting 4 from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and 9, the positive solution to the given equation is 9.

Q7

-419

The correct answer is -419. It's given that the expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a, b, and c are constants. Applying the distributive property to the given expression, $3x - 2319x + 6$, yields $3x19x + 3x6 - 2319x - 236$, which can be rewritten as $57x^2 + 18x - 437x - 138$. Combining like terms yields $57x^2 - 419x - 138$. Since this expression is equivalent to $ax^2 + bx + c$, it follows that the value of b is -419.

Q8

D

D. Day 12

Choice D is correct. The number of bacteria per milliliter is doubling each day. For example, from day 1 to day 2, the number of bacteria increased from 2.5×10^5 to 5.0×10^5. At the end of day 3 there are 10^6 bacteria per milliliter. At the end of day 4, there will be $10^6 \times 2$ bacteria per milliliter. At the end of day 5, there will be $(10^6 \times 2) \times 2$, or $10^6 \times (2^2)$ bacteria per milliliter, and so on. At the end of day d, the number of bacteria will be $10^6 \times (2^{d-3})$. If the number of bacteria per milliliter will reach 5.12×10^8 at the end of day d, then the equation $10^6 \times (2^{d-3}) = 5.12 \times 10^8$ must hold. Since 5.12×10^8 can be rewritten as 512×10^6, the equation is equivalent to $2^{d-3} = 512$. Rewriting 512 as 2^9 gives $d - 3 = 9$, so $d = 12$. The number of bacteria per milliliter would reach 5.12×10^8 at the end of day 12.

Choices A, B, and C are incorrect. Given the growth rate of the bacteria, the number of bacteria will not reach 5.12×10^8 per milliliter by the end of any of these days.

Q9

D

D. $\frac{1+\sqrt{11}}{2}$

Choice D is correct. A quadratic equation in the form $ax^2+bx+c=0$, where a, b, and c are constants, can be solved using the quadratic formula: $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$. Subtracting $2x+3$ from both sides of the given equation yields $2x^2-2x-5=0$. Applying the quadratic formula, where $a=2$, $b=-2$, and $c=-5$, yields $x=\frac{-(-2)\pm\sqrt{(-2)^2-4(2)(-5)}}{2(2)}$. This can be rewritten as $x=\frac{2\pm\sqrt{44}}{4}$. Since $\sqrt{44}=\sqrt{2^2(11)}$, or $2\sqrt{11}$, the equation can be rewritten as $x=\frac{2\pm2\sqrt{11}}{4}$. Dividing 2 from both the numerator and denominator yields $\frac{1+\sqrt{11}}{2}$ or $\frac{1-\sqrt{11}}{2}$. Of these two solutions, only $\frac{1+\sqrt{11}}{2}$ is present among the choices. Thus, the correct choice is D.

Choice A is incorrect and may result from a computational or conceptual error. Choice B is incorrect and may result from using $x=\frac{-b\pm\sqrt{b^2-4ac}}{a}$ instead of $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$ as the quadratic formula. Choice C is incorrect and may result from rewriting $\sqrt{44}$ as $4\sqrt{11}$ instead of $2\sqrt{11}$.

Q10 C

C. $(x+a)(x-a)$

Choice C is correct. Factoring –1 from the second, third, and fourth terms gives $x^2 - c^2 - 2cd - d^2 = x^2 - (c^2 + 2cd + d^2)$. The expression $c^2 + 2cd + d^2$ is the expanded form of a perfect square: $c^2 + 2cd + d^2 = (c + d)^2$. Therefore, $x^2 - (c^2 + 2cd + d^2) = x^2 - (c + d)^2$. Since $a = c + d$, $x^2 - (c + d)^2 = x^2 - a^2$. Finally, because $x^2 - a^2$ is the difference of squares, it can be expanded as $x^2 - a^2 = (x + a)(x - a)$.

Choices A and B are incorrect and may be the result of making an error in factoring the difference of squares $x^2 - a^2$. Choice D is incorrect and may be the result of incorrectly combining terms.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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